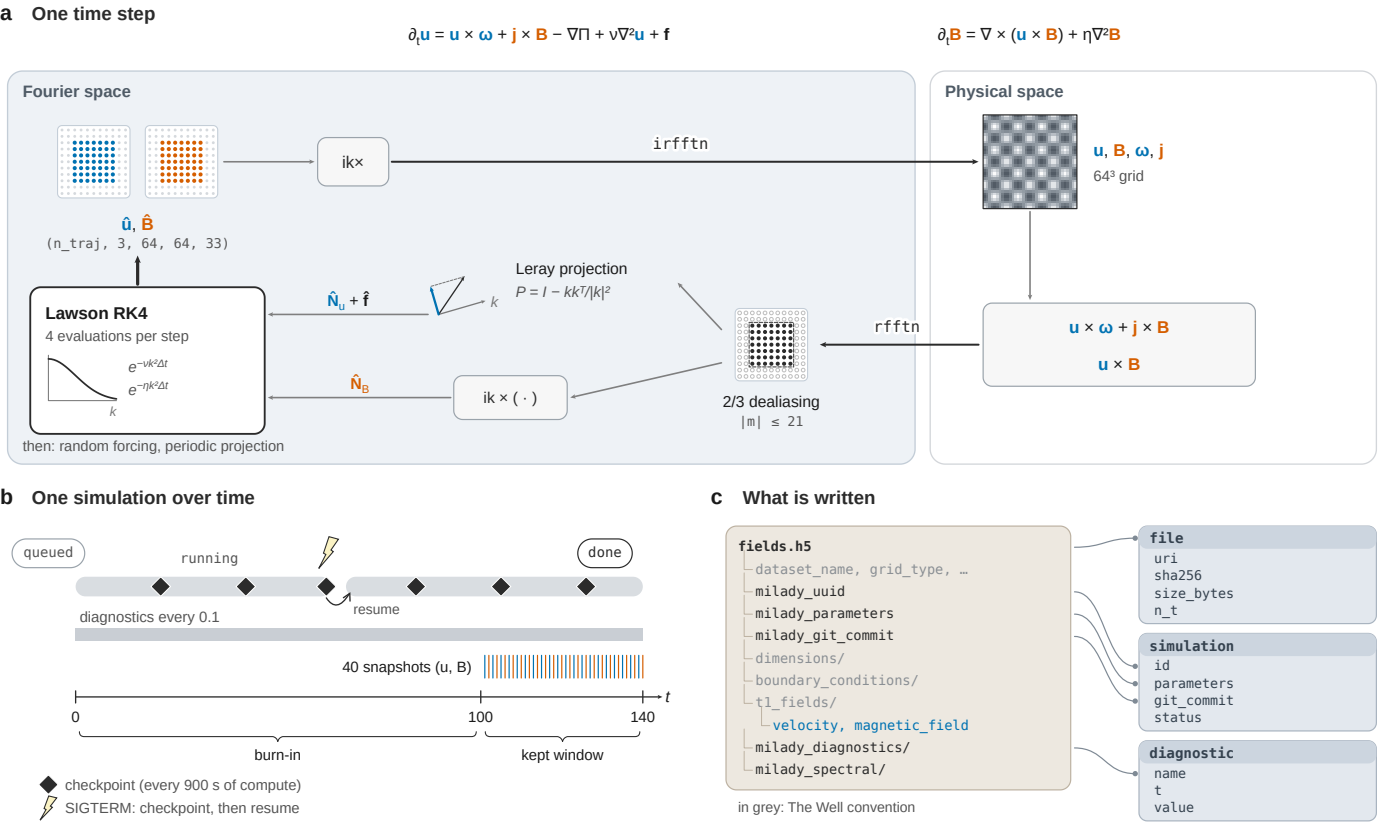

a One time step
$\partial_t \mathbf{u} = \mathbf{u} \times \boldsymbol{\omega} + \mathbf{j} \times \mathbf{B} - \nabla\Pi + \nu\nabla^2\mathbf{u} + \mathbf{f}$
$\partial_t \mathbf{B} = \nabla \times (\mathbf{u} \times \mathbf{B}) + \eta\nabla^2\mathbf{B}$
Fourier space
ik×
irfftn
Physical space
u, B, ω, j
64³ grid
û, B̂
(n_traj, 3, 64, 64, 33)
Lawson RK4
4 evaluations per step
$e^{-\nu k^2 \Delta t}$
$e^{-\eta k^2 \Delta t}$
k
then: random forcing, periodic projection
$\hat{\mathbf{N}}_u + \hat{\mathbf{f}}$
Leray projection
$P = I - kk^T/|k|^2$
$\hat{\mathbf{N}}_B$
ik × (·)
2/3 dealiasing
|m| ≤ 21
rfftn
u × ω + j × B
u × B
b One simulation over time
queued
running
done
resume
diagnostics every 0.1
40 snapshots (u, B)
t
0
100
140
burn-in
kept window
checkpoint (every 900 s of compute)
SIGTERM: checkpoint, then resume
c What is written
fields.h5
dataset_name, grid_type, …
milady_uuid
milady_parameters
milady_git_commit
dimensions/
boundary_conditions/
t1_fields/
velocity, magnetic_field
milady_diagnostics/
milady_spectral/
in grey: The Well convention
file
uri
sha256
size_bytes
n_t
simulation
id
parameters
git_commit
status
diagnostic
name
t
value